LUND UNIVERSITY

Centre for Mathematical Sciences
Mathematics, Faculty of Science

Written Examination
Number Theory
Wednesday January 9, 2019
Duration: 8.00–13.00

Solutions may of course be presented in Swedish! The sheets of the department should be used. Write your initials and at most one solution on each sheet handed in. Give careful motivations to your solutions!

1. Determine, using Legendre symbols, whether the congruences

$$x^2 + 7x + 17 \equiv 0 \pmod{283}, \quad \text{respectively,} \quad x^2 + 7x + 17 \equiv 0 \pmod{317}$$

are solvable or not. Note that 283 and 317 are primes.

2. Show that

$$\sum_{d|n} \mu^2(d) = 2^{\omega(n)},$$

where μ is Möbius μ-function and $\omega(n)$ denotes the number of distinct prime divisors of n.

3. Let p be a prime satisfying $p \equiv 5 \pmod 8$, and suppose that a is a quadratic residue modulo p. Show that one of the values

$$x = a^{\frac{p+3}{8}} \text{ or } x = 2a \cdot (4a)^{\frac{p-5}{8}}$$

is a solution to the congruence

$$x^2 \equiv a \pmod p.$$

4. For which prime powers p^k are there $x, y \in \mathbb{Z}$ such that $p \nmid x$, $p \nmid y$ and

$$x^2 + y^2 \equiv 0 \pmod{p^k}?$$

5. Let $x \in \mathbb{R} \setminus \mathbb{Q}$. Show that out of two consequtive convergents C_n, C_{n+1} to x, at least one satisfies

$$\left| x - \frac{a}{b} \right| < \frac{1}{2b^2}.$$

6. Let p be an odd prime, and let $a \in \mathbb{Z}$, $\gcd(a, p) = 1$. Show that

$$\sum_{j=0}^{p-1} (j^2 + a/p) = -1,$$

where $(j^2 + a/p)$ shall be interpreted as 0 in case $p \,|\, j^2 + a$.